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Research Project Report on the Topic:
Topological Analysis of Musical Genres and Popularity Using Audio Embeddings

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Аннотация

Персистентная гомология надёжно выявляет временную рекуррентную структуру в траекториях нейросетевых аудиоэмбедингов. Однако эта структура лишь слабо соответствует семантике музыкальных жанров, сильно зависит от выбора модели эмбедингов и локализована на повторяющихся музыкальных секциях лишь для меньшинства треков. В данной работе строится воспроизводимый пайплайн на датасете GTZAN (999 треков, 10 жанров) с четырьмя представлениями (MERT, MuQ, EnCodec, MIR) и тремя суррогатными контролями. Топологические признаки классифицируют жанр на уровне случая ($\text{macro-F1} \approx 0,22$), тогда как усреднённые эмбединги достигают 0,82, что демонстрирует разделение между топологией траекторий и жанровой семантикой. Работа выявляет методологические ловушки, включая почти двукратную переоценку доли «значимых» циклов при использовании коцикла вместо гомологического представителя. Вклад работы — карта применимости TDA к аудиоэмбедингам и строгий методологический шаблон; отрицательные результаты являются центральным вкладом.

Abstract

Persistent homology reliably detects temporal recurrence structure in trajectories of neural audio embeddings. However, this structure corresponds only weakly to musical genre semantics, depends strongly on the choice of embedding model, and is localized on repeated musical sections only for a minority of tracks. On the GTZAN dataset (999 tracks, 10 genres), a reproducible pipeline is built over four representations (MERT, MuQ, EnCodec, MIR) with a ladder of three surrogate controls. Topological features classify genre near chance level (macro-F1 ≈ 0.22), whereas time-averaged embeddings reach 0.82, exposing a sharp separation between trajectory topology and genre-level semantic organization. The work also identifies several methodological pitfalls, including an almost twofold overestimation of the share of “meaningful” loops caused by treating cocycle support as a geometric loop representative. The contribution is a map of where TDA is applicable to audio embeddings and a rigorous methodological template with controls, effect sizes, and reproducibility mechanisms; the negative results are a central contribution rather than a failure.

Keywords: topological data analysis, persistent homology, neural audio embeddings, music information retrieval, genre classification, surrogate controls, Takens embedding, persistence diagrams, Betti curves, GTZAN.

1 Introduction

Musical audio is a temporal signal: a track unfolds over time, and its content is determined not only by the set of timbres and instruments, but also by how they follow, return, and transform over time. Modern neural models encode audio as sequences of high-dimensional frame-level embeddings. Yet most downstream methods aggregate this sequence into a single vector, for example by mean pooling, max pooling, or weighted averaging. Such aggregation inevitably removes temporal structure.

An alternative view is to treat a track as a **trajectory** in feature space: a geometric object whose shape may carry information about musical content. This view is motivated by topological data analysis (TDA), especially persistent homology, which extracts shape invariants from point clouds (loops, connected components, voids) and summarizes them by persistence diagrams, which are stable under small perturbations [6].

Central question of the work: does the topological organization of neural audio embedding trajectories carry meaningful information about music, and if so, what kind of information?

It is important to distinguish two claims from the outset. *The topology of embedding trajectories* is what is measured here: geometric invariants of paths in a particular feature space. *The topology of music* would be a much stronger claim: that these invariants are properties of music itself rather than artifacts of a chosen model. This work studies the former and deliberately avoids claiming the latter.

The motivation for testing genre signal is concrete. If genres systematically differ in the density of structural repetitions (for example, pop and rock with verse–chorus forms versus jazz with improvisation), then their embedding trajectories might show different recurrent geometry, detectable through H_1 . This is a testable hypothesis, not a guaranteed outcome, and its failure in the final results is itself informative.

The key methodological choice is that the objective is **not to develop a new method**, but to **rigorously assess the limits of applicability of TDA** to audio embeddings: with explicit controls against self-deception, with effect sizes rather than reliance on p -values, and with an honest account of both positive and negative results. Negative results are treated as a central contribution, not as a failure.

The rest of the report is organized as follows. Section 2 reviews related work on TDA, audio embeddings, and delay embeddings. Section 3 formulates the hypotheses tested in this work. Section 4 describes the datasets. Section 5 details the full pipeline from embedding extraction through surrogate controls to classification and cycle analysis. Section 6 presents the experimental results for each hypothesis. Section 7 discusses the findings, methodological lessons, and limitations. Section 8 summarizes the conclusions, and Section 9 outlines directions for future work.

2 Literature Review

2.1 Topological Data Analysis and Persistent Homology

Persistent homology is a central tool in TDA. For a point cloud, a simplicial complex such as Vietoris–Rips is built as the radius grows, and topological features (connected components H_0 , loops H_1 , voids H_2) are born and die at certain thresholds. A persistence diagram is a multiset of birth–death pairs and provides a stable descriptor of data shape. The theoretical foundations and stability properties are developed in the work of Edelsbrunner and Harer, Chazal et al. [6], and related literature. Persistence images [10], Betti curves, and persistence landscapes [7] provide stable vector representations suitable for machine learning.

2.2 TDA for Audio and Music

TDA has previously been applied to music mainly in settings involving self-similarity matrices, chroma features, or symbolic representations. Classical approaches often analyze temporal signals directly through sliding-window persistence [16] for periodicity in time series, or measure similarity and repetitions through chroma representations [17].

Recent advances include applying persistent homology to log-mel spectrograms for robust audio fingerprinting [22], and analyzing symbolic music graphs. However, there is a significant research gap: as of mid-2026, there are no published works applying persistent homology directly to the latent trajectories of modern deep music foundation models (e.g., MERT, MuQ) for musical similarity or genre analysis. Applying TDA directly to trajectories of neural embeddings, rather than to raw signals or self-similarity matrices, is thus an open direction.

2.3 Why Frame-Level Embeddings Are Required for TDA

A crucial distinction for the present work is between *clip-level embeddings* and *frame-level embedding trajectories*. In the landscape of deep audio embeddings, many models are optimized for tagging or contrastive retrieval (e.g., VGGish [24], YAMNet, CLAP [23], MuLan). These architectures typically output a single pooled vector per clip, most often via global mean-pooling or a CLS token.

While highly effective for semantic similarity and zero-shot classification, global pooling operations completely destroy the temporal locality and moment-to-moment dynamics of the track. Since persistent homology in this context is explicitly applied to the shape of a track’s path in feature space to find recurrent geometry, the representation must preserve temporal order and output a sequence of vectors. For TDA, mean-pooling is a destructive operation that collapses a trajectory in $\mathbb{R}^{T \times D}$ into a single centroid in $\mathbb{R}^D$, erasing the very loops that H_1 homology aims to detect.

2.4 Neural Audio Embeddings Used in This Work

To explore the topology of trajectories, this work specifically selects representations that provide frame-level sequences, covering three fundamentally different inductive biases: self-supervised semantic learning, codec reconstruction, and hand-crafted signal processing.

MERT [1] is a widely used acoustic music understanding model based on a BERT-style Transformer encoder. It provides high-resolution 75 Hz frame-level hidden states, making it a strong music-specific foundation baseline.

MuQ [2] is a newer Conformer-based music SSL model trained with a Mel Residual Vector Quantization (Mel-RVQ) objective. It outputs 25 Hz embeddings and provides a modern, computationally efficient alternative to MERT with a different temporal resolution and architecture.

EnCodec [3] is not a semantic SSL model, but a neural audio codec. The continuous latent space before quantization provides a compact 128-dimensional representation at 75 Hz. Its inclusion tests whether topological structures arise not only from semantic music training but also from pure acoustic reconstruction geometry.

MIR features (librosa [20]) serve as a classical, interpretable low-dimensional baseline (chroma and spectral features). It preserves explicit frame-level structure and helps determine if topological findings are exclusive to deep foundation models or exist in classical descriptors.

This selection is intentionally heterogeneous. If topological effects were observed only in MERT or MuQ, they could be interpreted as artifacts of a specific SSL objective. By including EnCodec and MIR, the work tests whether recurrent topology is a property of semantic music representations, reconstruction-oriented audio latents, or more general frame-level audio descriptors.

2.5 Layer Choice in Transformer Audio Models

For Transformer-based models, the choice of the extracted layer significantly impacts the representation’s geometry. Intermediate layers were preferred over final pooled outputs because Transformer layers encode different levels of abstraction.

Evaluations on benchmarks like MARBLE [25] show that early layers tend to preserve local acoustics, and later layers specialize heavily toward acoustic tokens or pretraining objectives. Middle layers (e.g., 5–9 for 95M-parameter models, or 13–18 for 330M-parameter models) provide the best compromise for extracting musically relevant properties such as pitch, harmony, and structural context. Therefore, MERT layer 7 and MuQ layer 10 were deliberately chosen as contextual frame-level representations that still preserve the temporal trajectory needed for Takens embedding and subsequent topological analysis.

2.6 Delay Embedding and Takens’ Theorem

Takens’ theorem [11] motivates the use of delay embeddings for time series: under certain conditions, such an embedding preserves the structure of the phase space up to a smooth transformation. Further theoretical developments include the embedology framework [12] and practical methods for select-

ing embedding parameters via Average Mutual Information (AMI) [13] and False Nearest Neighbors (FNN) [14, 15].

However, classic parameter selection methods like AMI and FNN were originally formulated for reconstructing phase spaces from scalar (1D) time series. In the context of modern neural audio embeddings, the "observable" is already a dense, high-dimensional vector (e.g., up to 1024 dimensions, or 32 dimensions after PCA). Applying FNN to a sequence of such high-dimensional vectors to find an optimal embedding dimension becomes computationally prohibitive and conceptually mismatched. In extremely high-dimensional spaces, the curse of dimensionality renders spatial neighborhoods sparse, making the ratio of false to true nearest neighbors highly unstable and uninformative. Consequently, this work treats the sliding window approach not as a strict chaotic phase space reconstruction, but rather as a geometric heuristic to convert the temporal order of high-dimensional frames into point-cloud geometry. The delay parameters are therefore chosen empirically, and the validity of the resulting topological structures is verified against robust surrogate controls rather than intrinsic phase-space metrics.

At the same time, Takens-style embeddings naturally tend to produce recurrent and loop-like geometry for periodic or quasiperiodic signals, not only for music. Therefore, the mere presence of H_1 loops in a delay-embedded cloud is not evidence of specifically musical structure. The loops must be compared against surrogate controls [18] that destroy different properties of the time series. This theoretical consideration directly motivates the surrogate ladder used in the work.

3 Problem Statement and Hypotheses

Formally, each track is represented as a sequence of frame-level embeddings $\{e_t\}_{t=1}^T$, where $e_t \in \mathbb{R}^d$. From this sequence, a point cloud $X \subset \mathbb{R}^D$ is built using Takens embedding, after which persistent homology $\text{PH}(X)$ is computed. The goal is to assess to what extent descriptors of $\text{PH}(X)$ carry information about the musical content of the track.

The following hypotheses were tested.

H1 — Reality of loops. The observed H_1 loops in embedding trajectories are not computational artifacts: their persistence significantly exceeds three surrogate controls that test the contribution of temporal order, linear-spectral smoothness, and the empirical distribution of the series.

H2 — Genre specificity of topology. Topological trajectory descriptors carry information about genre: the mean Wasserstein distance between diagrams of tracks from the same genre is smaller than the mean distance between tracks from different genres.

H3 — Nonlinear temporal structure. The observed loops contain a component beyond the chosen linear-spectral null control: the persistence of real trajectories exceeds the phase surrogate, which preserves the spectral structure of the series but destroys the original phase organization. This test is not a strict proof of nonlinearity in the dynamical-systems sense, and it does not test “music versus non-music”; the latter would require comparison against non-musical audio, which is outside the scope of the work.

H4 — Consistency of topological organization across representations. Different audio embeddings reveal similar topological organization: pairwise Wasserstein distance matrices between representation spaces are correlated, tested via the Mantel test.

H5 — Practical usefulness. Topological descriptors are useful for genre classification: they outperform surrogate controls and complement aggregated embeddings.

H-loop. Vertices of a representative homological cycle for the most persistent loop show increased chromatic similarity, i.e. a topological loop partially corresponds to harmonically repeated sections of a track.

4 Data

GTZAN. The main dataset [4] contains 999 tracks (one defective track excluded), 10 genres (blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, rock), approximately 100 tracks per genre, and 30 seconds per track. GTZAN is a standard benchmark in the literature, although it is known to contain label errors and duplicates [5]; this is taken into account when interpreting the results.

Split: 80% train and 20% test, stratified by genre. All preprocessing components (StandardScaler, PCA) are fitted only on the training part and then applied to the test part without data leakage.

H-loop branch. For testing H-loop, a separate collection of 199 Spotify tracks was used (central 90 seconds), with pre-annotated repetition structure. Of these, 192 tracks yielded a valid test.

Popularity branch. For an exploratory popularity analysis, `top50_tracks.csv` was used, which contains Spotify popularity metadata. Only tracks with already computed 90-second MuQ and MIR embeddings were analyzed: 198 matched tracks from four genres (electronic, hip-hop, pop, reggae). This is not a representative popularity dataset: it is restricted to already popular tracks and therefore has a truncated popularity range. The analysis is interpreted only as a weak exploratory check of whether topological loop statistics covary with Spotify popularity inside this biased subset.

5 Methodology

The pipeline is a sequence of mathematical transformations of the same object: an audio track is first converted into a time series of features, then into a point cloud, then into a persistence diagram, and finally into statistics, distances, classification features, and geometric cycle representatives:

$$\text{audio} \rightarrow E \rightarrow X \rightarrow Z \rightarrow D_1 \rightarrow \{\text{max-persistence}, W(D_i, D_j), \beta_D, \text{cycle vertices}\}. \quad (1)$$

Here E is the original frame-level feature matrix, X is the normalized and dimension-reduced time series, Z is the Takens point cloud, and D_1 is the persistence diagram for one-dimensional homology (see Figure 1).

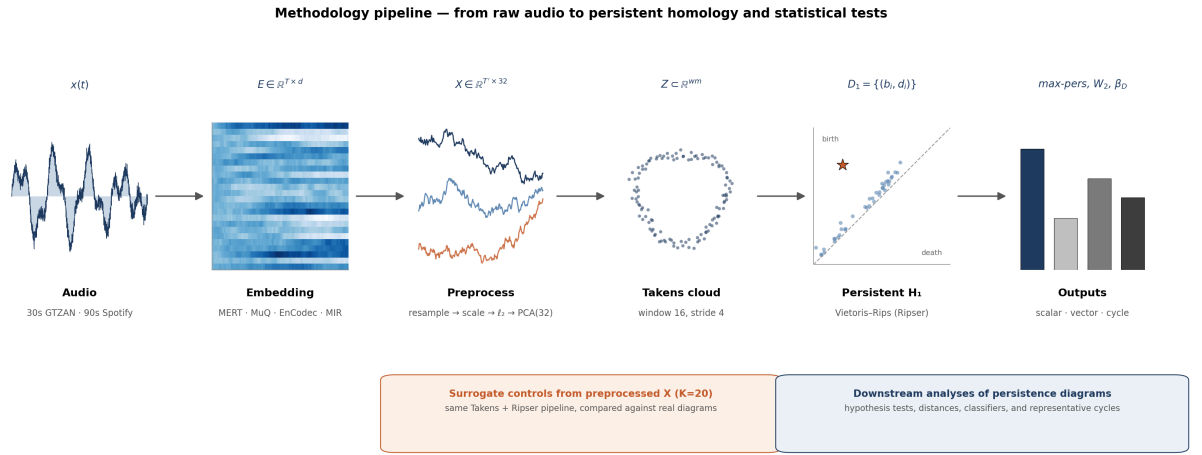


Figure 1 – Overview of the topological pipeline.

5.1 Embedding Extraction

Four sources of features were used; they differ in architecture and frame rate:

- **MERT** [1] (layer 7): a self-supervised acoustic masked-prediction model, 75 frames/s, dimension 768.
- **MuQ** [2] (layer 10): a music SSL model, native 25 frames/s, dimension 1024.
- **EnCodec** [3] (continuous latent): a neural audio codec, 75 frames/s, dimension 128. The continuous latent before quantization was used, not discrete RVQ codes.
- **MIR** (librosa [20]): traditional frame-level features: chroma_cens (12 pitch classes) plus spectral centroid, flatness, and bandwidth; approximately 47 frames/s, dimension 15.

Embeddings were extracted once and cached to disk. For GTZAN (999 tracks), the cache size was several gigabytes.

5.2 Preprocessing and Common Denominator

For fair cross-comparison, all representations were brought to a common denominator. The order of steps is essential.

1. **Resampling of the frame sequence** to a common $f_t = 25$ frames/s using the Fourier method. This removes frame-rate differences between MERT/EnCodec (75 fps) and MuQ (25 fps). If the original frequency is f_s , the new length is

$$T' = \text{round}\left(T \frac{f_t}{f_s}\right). \quad (2)$$

2. **StandardScaler** (per feature, fitted on train): feature-scale alignment. For coordinate j :

$$S(x)_j = \frac{x_j - \mu_j}{\sigma_j + \varepsilon}, \quad \mu_j = \frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} x_j, \quad \sigma_j^2 = \frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} (x_j - \mu_j)^2, \quad (3)$$

where $\mathcal{T}$ is the set of all training frames for a representation.

3. **Unit-sphere normalization of each frame**:

$$\tilde{x} = \frac{S(x)}{\|S(x)\|_2}. \quad (4)$$

This reduces the influence of norm/energy. For two normalized frames $\tilde{x}, \tilde{y}$, Euclidean distance becomes equivalent to angular distance: $\|\tilde{x} - \tilde{y}\|_2^2 = 2 - 2 \cos(\tilde{x}, \tilde{y})$.

4. **PCA to 32 components** (fitted on train): dimensionality reduction before Takens embedding. If C is the covariance matrix of normalized training frames, the first m eigenvectors $u_1, \dots, u_m$ are selected, and the final frame is

$$R(x) = (\langle \tilde{x}, u_1 \rangle, \dots, \langle \tilde{x}, u_m \rangle). \quad (5)$$

In the main runs, $m = 32$; if the original dimension is less than 32, $m = \min(d, 32)$. For MIR, where the original dimension is $15 < 32$, PCA is not applied. Explained variance: EnCodec $\approx 84\%$, MERT $\approx 36\%$, MuQ $\approx 22\%$ (the low value for MuQ reflects high dimensionality and a more uniform variance distribution).

The parameters μ_j , σ_j , and u_j are estimated only on the training split and then applied unchanged to all tracks, preventing test tracks from influencing the coordinate system in which their topology is measured.

5.3 Point Cloud Construction: Takens Embedding

After preprocessing, each track is represented as $X = (x_1, \dots, x_{T'})^\top$, $x_t \in \mathbb{R}^m$. From this sequence, a sliding-window Takens embedding is built with window length $w = 16$ frames and stride $s = 4$ frames.

Each cloud point is a short trajectory of w consecutive frames flattened into one vector:

$$z_j = (x_j, x_{j+1}, \dots, x_{j+w-1}) \in \mathbb{R}^{wm}, \quad j = 1, 1+s, 1+2s, \dots, T'-w+1. \quad (6)$$

The number of points is $N = \lfloor (T' - w)/s \rfloor + 1$. For a 30-second GTZAN track at 25 frames/s, $T' \approx 750$ and $N \approx 184$. In the main experiments, the upper bound is $N_{\max} = 300$, so GTZAN clouds are effectively not subsampled. In the H-loop branch for 90-second fragments, a mapping from cloud vertex to audio time is additionally stored:

$$t(v) = \frac{\text{start}(v) + w/2}{25}. \quad (7)$$

Intuitively, temporal order is converted into point-cloud geometry; when a trajectory returns to a nearby region of feature space, this can appear as an H_1 loop. However, Takens embeddings naturally tend to generate recurrent geometry for quasiperiodic signals; therefore, the presence of loops alone proves nothing. What matters is whether real loops exceed surrogate controls.

5.4 Computing Persistent Homology

Persistent homology is computed through the Vietoris–Rips filtration using Ripser [8] (maxdim = 1, i.e. H_0 and H_1). As ε grows, H_1 classes are born and die, producing a diagram:

$$\text{VR}_\varepsilon(Z) = \left\{ \sigma \subseteq Z : \max_{u,v \in \sigma} \|u - v\|_2 \leq \varepsilon \right\}, \quad D_1(Z) = \{(b_i, d_i)\}_{i=1}^r. \quad (8)$$

The main scalar feature for controls is maximum H_1 persistence:

$$\text{mp}(Z) = \max_{(b_i, d_i) \in D_1(Z)} (d_i - b_i). \quad (9)$$

Examples of such diagrams for real tracks and their surrogate controls are shown in Figure 2.

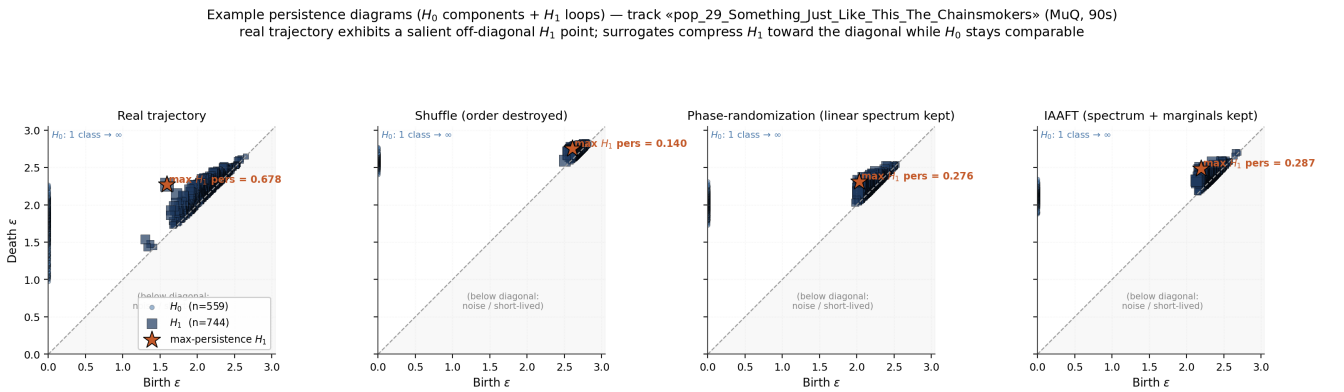


Figure 2 – Examples of persistence diagrams for real audio embeddings and their surrogate controls.

To interpret Figure 2, note that the horizontal axis represents the birth time (the filtration radius ε at which a topological loop is formed) and the vertical axis represents the death time (the radius at which the loop is filled in). Points located close to the diagonal line ($y = x$) represent short-lived loops, which are typically interpreted as topological noise or minor local fluctuations. Conversely, points located far

above the diagonal correspond to highly persistent loops (large $d_i - b_i$), indicating robust, large-scale recurrent geometry in the embedding trajectory. As seen in the example, the persistence diagram for the real audio track exhibits points noticeably farther from the diagonal than its surrogate controls (Shuffle, Phase, and IAAFT). This visual separation intuitively demonstrates how the controls destroy the genuine long-term temporal structure of the track, resulting in diagrams dominated by noise clustered near the diagonal.

Two tracks are compared by Wasserstein distance between diagrams:

$$W_p(D, D') = \left(\inf_{\gamma} \sum_{x \in D} \|x - \gamma(x)\|_{\infty}^p \right)^{1/p}, \quad (10)$$

where matching diagram points to the diagonal is allowed. From the distance matrix $A_{ij} = W(D_i, D_j)$, within- and between-genre means are computed, and the H2 statistic is $\Delta = \bar{W}_{\text{between}} - \bar{W}_{\text{within}}$. Confidence intervals for Δ are computed by bootstrapping tracks, not pairs.

5.5 Diagram Vectorization

To pass topological features into classical ML models, persistence diagrams must be vectorized. A persistence image [10] was initially considered: a weighted Gaussian smoothing of the diagram on a regular grid. However, for diagrams dominated by short-lived loops (points near the diagonal), persistence images degenerated into nearly homogeneous matrices and lost the informative signal.

After this diagnosis, the **Betti curve** was selected. For each filtration threshold τ_{ℓ} on a grid ($q = 20$ points in the main experiments), the number of live H_1 classes is counted:

$$\beta_D(\tau_{\ell}) = \sum_i \mathbf{1}\{b_i \leq \tau_{\ell} < d_i\}, \quad \ell = 1, \dots, q. \quad (11)$$

The grid range is fixed on the training part, so train/test and real/surrogate vectors lie in the same coordinate system. The Betti curve proved empirically more informative than persistence images under the chosen grid and parameters, although it does not itself suppress short-lived classes and still requires control of the filtration range.

5.6 Surrogate Controls

The core of the methodology is a **ladder of three surrogate controls**, each destroying a different property of the signal. The surrogate is applied before Takens embedding, i.e. to the time series X , not to the already constructed point cloud Z . For each control type, $K = 20$ realizations are generated.

1. **Shuffle**: frames of the original sequence are randomly permuted before Takens embedding:

$$X^{\text{shuf}} = (x_{\pi(1)}, \dots, x_{\pi(T)}). \quad (12)$$

This destroys temporal order entirely. Null hypothesis: the observed topology is explained only by the distribution of frames, not their order.

2. **Phase randomization:** for each frequency component, the DFT is taken, amplitudes are preserved, and phases are randomized. For a multivariate sequence $X \in \mathbb{R}^{T \times m}$, one random phase ϕ_ω per frequency ω is shared across all PCA components:

$$\widehat{X}_j^{\text{rand}}(\omega) = \widehat{X}_j(\omega) e^{i\phi_\omega}, \quad j = 1, \dots, m, \quad \phi_\omega \sim U(0, 2\pi). \quad (13)$$

The DC component and Nyquist frequency are fixed so the inverse transform remains real. Because the same multiplier $e^{i\phi_\omega}$ is applied to all coordinates, the cross-spectral matrix is preserved. Null hypothesis: the topology is explained by linear-spectral smoothness.

3. **IAAFT** [18] (Iteratively Amplitude-Adjusted Fourier Transform): preserves both the amplitude spectrum and the empirical distribution of values, componentwise. At each iteration, the current signal y first receives the target spectrum:

$$y \leftarrow \mathcal{F}^{-1} \left(|\hat{x}| \frac{\hat{y}}{|\hat{y}|} \right), \quad (14)$$

then its values are replaced by rank matching to the original sorted values. Iterations stop by relative spectral error or at the limit of 200 iterations. Null hypothesis: the topology is explained by preserved componentwise spectrum and distribution.

The three null hypotheses are not strictly nested. The phase surrogate preserves cross-component spectral structure; IAAFT does not, because it is componentwise, but it preserves empirical marginal distributions.

For each track, the null level is estimated by $\overline{\text{mp}}_i^{(c)} = \frac{1}{K} \sum_{k=1}^K \text{mp}(D_i^{(c,k)})$, and the track-level difference $\delta_i^{(c)} = \text{mp}(D_i) - \overline{\text{mp}}_i^{(c)}$ is used for the paired Wilcoxon test and rank-biserial effect size.

5.7 Statistical Framework

Effect sizes first, p -values second. With 999 tracks, there are approximately 500 000 diagram pairs; at this scale, p -values such as 10^{-165} are a warning sign, not a triumph. The primary reported quantities are rank-based effect sizes: rank-biserial correlation (Cliff's δ).

Track-level bootstrap. Diagram pairs are dependent because each track appears in many pairs. Pair-level bootstrap underestimates confidence intervals. The correction is aggregation to track-level statistics followed by bootstrap over tracks (1000 iterations).

Mantel test. To compare pairwise distance matrices between representations, rows and columns are permuted synchronously, rather than simply correlating upper triangles under invalid elementwise shuffling.

FDR correction [19] (Benjamini–Hochberg) is applied within meaningful test families: `controls_shuffle` (4 tests), `controls_persistence` (8), `mantel` (3), and `chroma_loop` (192 tracks).

Pseudoreplication is removed: in the shuffle test, aggregation by track before Wilcoxon prevents treating K surrogate realizations as independent observations.

Permutation test for H-loop: instead of an unpaired Mann–Whitney test on dependent pairs, a permutation test is used over random time-point sets within each track.

5.8 Downstream Classification

To assess practical usefulness (H5), genre classification was performed on the 10 GTZAN genres. Two classifiers were used:

- **LogisticRegression** (L2): a linear model for assessing linear separability of features [21].
- **HistGradientBoostingClassifier** (HGB): a nonlinear model. It tests whether topological features become practically useful under a more expressive classifier; if performance remains near chance or comparable to surrogates, the “weak linear classifier” loophole is closed.

Six feature sets were used: `topology` (Betti curves of real data); `topology@shuffle`, `topology@phase`, `topology@IAAFT` (Betti curves of surrogates, averaged over K realizations); `mean_pool` (time average of the preprocessed sequence); and `concat` (concatenation of `topology` and `mean_pool`).

Performance is evaluated on a fixed stratified test split by macro-F1:

$$\text{macroF1} = \frac{1}{G} \sum_{g=1}^G \frac{2 \text{precision}_g \text{recall}_g}{\text{precision}_g + \text{recall}_g}, \quad (15)$$

with bootstrap confidence intervals (1000 resamples, 95%).

5.9 Loop Localization Analysis

For H-loop, the most persistent H_1 class is identified. Two types of representative are distinguished:

- **Cocycle** (Ripser): the standard Ripser output is a representative cocycle in cohomology. Its vertex support is not a geometric loop contour. On real tracks, median support is 85 points out of 300; on a reference circle it covers about 60% of points.
- **Cycle** (Dionysus [9]): a representative cycle in homology, computed with Dionysus. It is not canonical, shortest, or optimal, but is about twice as compact as cocycle support: median 36 points out of 300.

An etalon comparison of these representatives on a circle is shown in Figure 3.

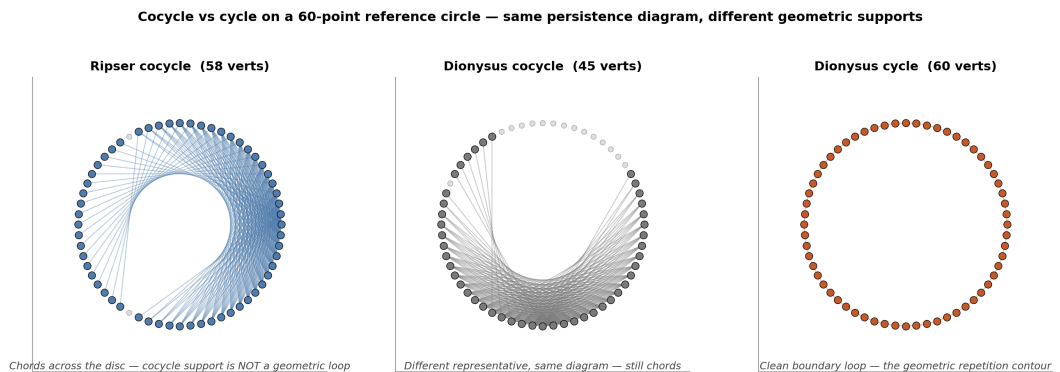


Figure 3 – Etalon comparison of cycle and cocycle representatives on a circle.

To evaluate the chromatic correspondence of the topological loops, Chroma Energy Normalized Statistics (CENS) features are computed using the `librosa` library. The extraction process begins with calculating a standard chromagram $C(t, p)$, representing the energy of pitch class $p \in \{1, \dots, 12\}$ at frame t . To achieve invariance to local dynamics and loudness, the vectors are first ℓ_1 -normalized:

$$C_{\text{norm}}(t, p) = \frac{C(t, p)}{\sum_{q=1}^{12} C(t, q)}, \quad (16)$$

where $C(t, p)$ is the energy of pitch class p at frame t .

Subsequently, the normalized values $C_{\text{norm}}(t, p)$ are quantized using a step function $Q(\cdot)$ to reduce the influence of noise, subtle timbral variations, and minor performance differences. The quantized sequence is then smoothed over a temporal window of length w to capture broader harmonic progressions rather than isolated notes. Finally, the smoothed vectors are ℓ_2 -normalized to produce the final CENS feature vector $c(t)$ whose p -th component is:

$$c(t)_p = \frac{\sum_{k=t-w/2}^{t+w/2} Q(C_{\text{norm}}(k, p))}{\sqrt{\sum_{q=1}^{12} \left(\sum_{k=t-w/2}^{t+w/2} Q(C_{\text{norm}}(k, q)) \right)^2}}, \quad (17)$$

where w is the smoothing window length, and $Q(\cdot)$ denotes the quantization step function. This ℓ_2 -normalization is crucial because it ensures that the Euclidean distance between two vectors is equivalent to their angular (cosine) distance, providing a robust and mathematically justified metric for chromatic similarity.

Cycle vertices are mapped back to audio time. The test statistic is the mean cosine similarity between cycle vertices separated by at least 5 seconds:

$$S_{\text{cycle}} = \frac{1}{|Q|} \sum_{(u,v) \in Q} \frac{\langle c(t_u), c(t_v) \rangle}{\|c(t_u)\| \|c(t_v)\|}, \quad |t_u - t_v| \geq 5 \text{ s}. \quad (18)$$

The null distribution is generated by selecting the same number of random time points, and the p -value uses the +1 correction: $p = \frac{1 + \#\{S_{\text{null}} \geq S_{\text{cycle}}\}}{1 + R}$. An example of a chromagram with localized cycle vertices is presented in Figure 4.

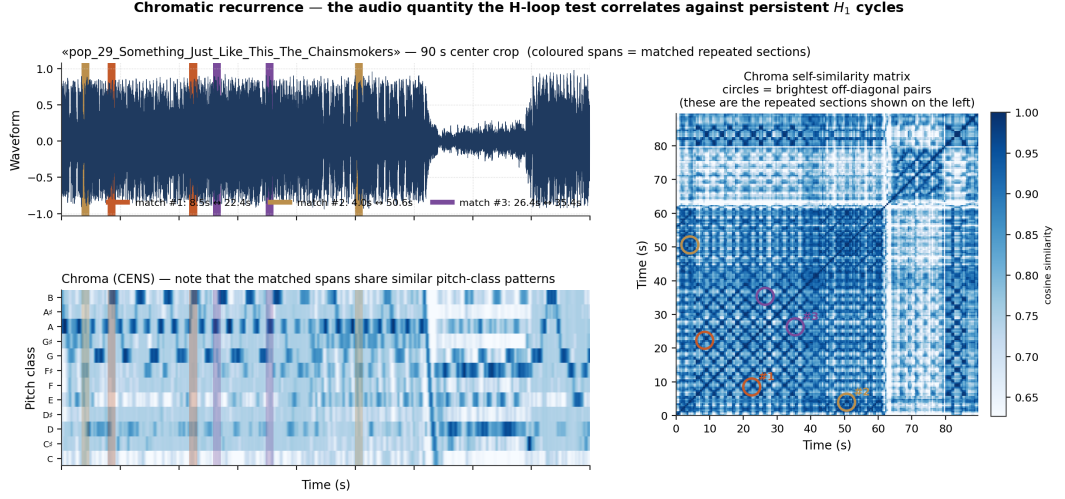


Figure 4 – Example of a chromagram with localized topological cycle vertices.

5.10 Popularity Analysis

For the popularity branch, each cached Spotify90 track is matched to metadata by genre and rank. For each representation space $s \in \{\text{MuQ}, \text{MIR}\}$, the H-loop statistics are reused: maximum H_1 persistence $p_i^{(s)}$, the number of vertices in the representative cycle $|V_i^{(s)}|$, the temporal span $\Delta t_i^{(s)}$, and chroma similarities for the cycle and null model. The main derived statistic is chromatic excess $e_i^{(s)} = \bar{c}_{\text{cycle},i}^{(s)} - \bar{c}_{\text{null},i}^{(s)}$. The relation to Spotify popularity q_i is evaluated by Spearman correlation, both raw and genre-centered:

$$\tilde{q}_i = q_i - \bar{q}_{g(i)}, \quad \tilde{m}_i^{(s)} = m_i^{(s)} - \bar{m}_{g(i)}^{(s)}. \quad (19)$$

A high-vs-low comparison uses lower and upper tertiles and a two-sided Mann–Whitney test. These tests are exploratory.

5.11 Ablation Checks and Reproducibility

Robustness is tested as a set of one-factor perturbations of the same mathematical pipeline. In these checks the operator

$$\mathcal{P}_\theta : E_i \mapsto X_i^\theta \mapsto Z_i^\theta \mapsto D_i^\theta$$

is recomputed under a single changed factor θ , while all other choices are held fixed. The reported ablations are run on a stratified subset of approximately 100 GTZAN tracks, 10 per genre. For each variant, the same key quantity is recomputed, for example:

$$E_\theta = \text{rankBiserial} \left(\{\text{mp}(D_i^\theta)\}_i, \{\overline{\text{mp}}(D_{i,\text{shuffle}}^\theta)\}_i \right), \quad (20)$$

and the spread of E_θ across variants is compared.

The ablation grid covers the following factors. First, resampling is tested by comparing the baseline $f_t = 25$, $w = 16$, $s = 4$ with a no-downsampling variant for native 75 fps representations:

$$\frac{16}{25} = \frac{48}{75} = 0.64 \text{ s}, \quad \frac{4}{25} = \frac{12}{75} = 0.16 \text{ s}.$$

Thus the physical Takens window and hop are preserved while the frame-rate conversion itself is removed. A linear-interpolation resampler is also compared against the Fourier resampler at the same f_t, w, s . This specifically checks whether the MuQ advantage could be an artifact of downsampling MERT/EnCodec from 75 to 25 fps.

Second, PCA dimensionality is varied as $m \in \{16, 32, 64, 128\}$, which changes the Takens ambient dimension from $16m = 256$ to 2048. Third, four normalization regimes are compared:

$$x \mapsto S_{\text{std}}(x), \quad x \mapsto \frac{S_{\text{std}}(x)}{\|S_{\text{std}}(x)\|_2}, \quad x \mapsto \frac{S_{\text{rob}}(x)}{\|S_{\text{rob}}(x)\|_2}, \quad x \mapsto x,$$

where S_{std} is mean/std scaling and S_{rob} is median/IQR scaling. Fourth, Takens time scale is varied as $\ell \in \{0.32, 0.64, 1.28, 2.56\}$ seconds with

$$w_\ell = \text{round}(\ell f_t), \quad s_\ell = \text{round}(w_\ell/4).$$

This is evaluated both with fixed PCA dimension $m = 32$ and with an approximately fixed Takens ambient dimension using $m_\ell = \lfloor 512/w_\ell \rfloor$. Fifth, stride is varied at fixed window $w = 16$ as $s \in \{8, 4, 2\}$, changing the cloud density

$$N_s = \left\lfloor \frac{T' - w}{s} \right\rfloor + 1.$$

Sixth, split stability is checked over five seeds, so train-only standardization and PCA are refit on different stratified splits. Finally, the scalar persistence summary is varied: besides maximum lifetime, top-3 mean lifetime, total lifetime, number of loops with lifetime > 0.05 , persistence entropy, and Betti-curve area are evaluated. Distance concentration is monitored by

$$\kappa(Y) = \frac{\text{std}_{i < j} \|y_i - y_j\|_2}{\text{mean}_{i < j} \|y_i - y_j\|_2}$$

for raw frames, PCA-normalized frames, and Takens clouds.

Reproducibility is ensured by a fixed base seed $s_0 = 42$ and deterministic seeds at the track and repetition level:

$$\text{seed}_{\text{shuffle}} = 1000s_0 + 100i + k, \quad \text{seed}_{\text{phase}} = 2000s_0 + 100i + k, \quad \text{seed}_{\text{IAAFT}} = 3000s_0 + 100i + k, \quad (21)$$

where i is the track index and k the repetition index. PCA is computed with deterministic full SVD so the final coordinate system does not depend on batch order.

6 Results

6.1 H1 — Reality of Topological Loops

This is the strongest and most robust result of the work. The surrogate-control ladder is summarized in Table 1 and Figure 5.

Table 1 – Mean maximum H_1 persistence over tracks by channel and condition

Channel	Real	Shuffle	Phase	IAAFT
MERT	0.465	0.154	0.324	0.333
MuQ	0.326	0.111	0.233	0.228
EnCodec	0.374	0.213	0.266	0.263
MIR	1.595	0.323	1.248	1.267

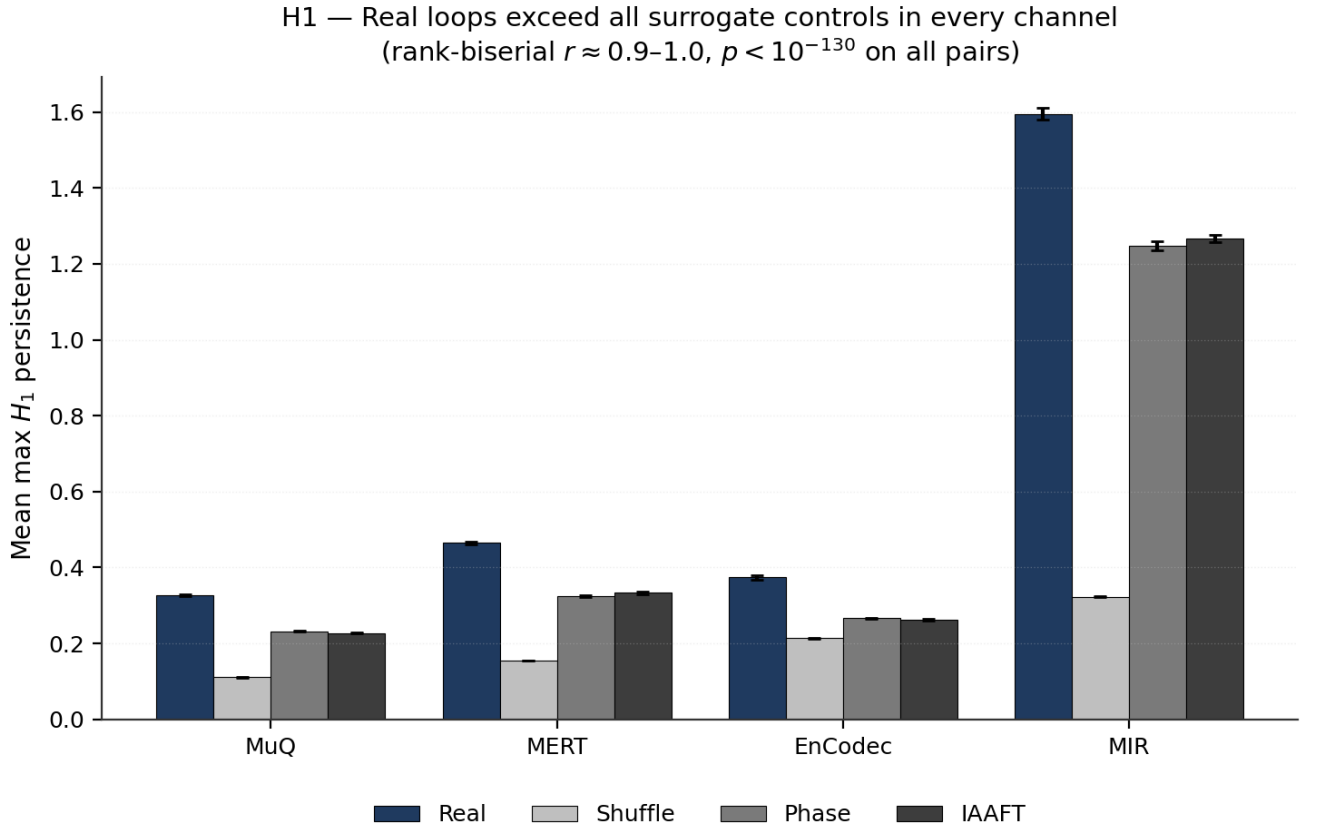


Figure 5 – Maximum H_1 persistence for real tracks compared to surrogate controls.

Rank-biserial effect sizes are shown in Table 2.

All tests have $p < 0.001$ (actual p -values are around 10^{-130} to 10^{-165} ; they are reported as $p < 0.001$ because such extreme values mostly reflect large sample size).

Conclusion: H_1 loops in embedding trajectories do not reduce to point order destruction (shuffle), the chosen linear-spectral surrogate (phase), or componentwise preserved spectrum and distribution (IAAFT). The topology detects robust recurrent structure. The important caveat is that this result characterizes trajectories in particular embedding spaces, not “the topology of music” as such.

Table 2 – Rank-biserial effect sizes for real vs. surrogate comparisons across all four channels

Comparison	Effect size (all 4 channels)
Real vs. shuffle	0.998–1.000 (large)
Real vs. phase	0.90–0.99 (large)
Real vs. IAAFT	0.89–0.99 (large)

6.2 H2 — Within-/Between-Genre Distances

The gap is defined as $\Delta = \bar{W}_{\text{between}} - \bar{W}_{\text{within}}$, with track-level 95% bootstrap CI (see Table 3 and Figure 6).

Table 3 – Within- and between-genre Wasserstein distances with genre gap

Channel	Within	Between	Gap	95% CI
MuQ	13.55	16.16	2.60	[2.38, 3.13]
EnCodec	14.55	16.12	1.56	[1.34, 2.11]
MIR	17.23	18.50	1.27	[1.21, 1.72]
MERT	14.85	16.04	1.19	[1.08, 1.61]

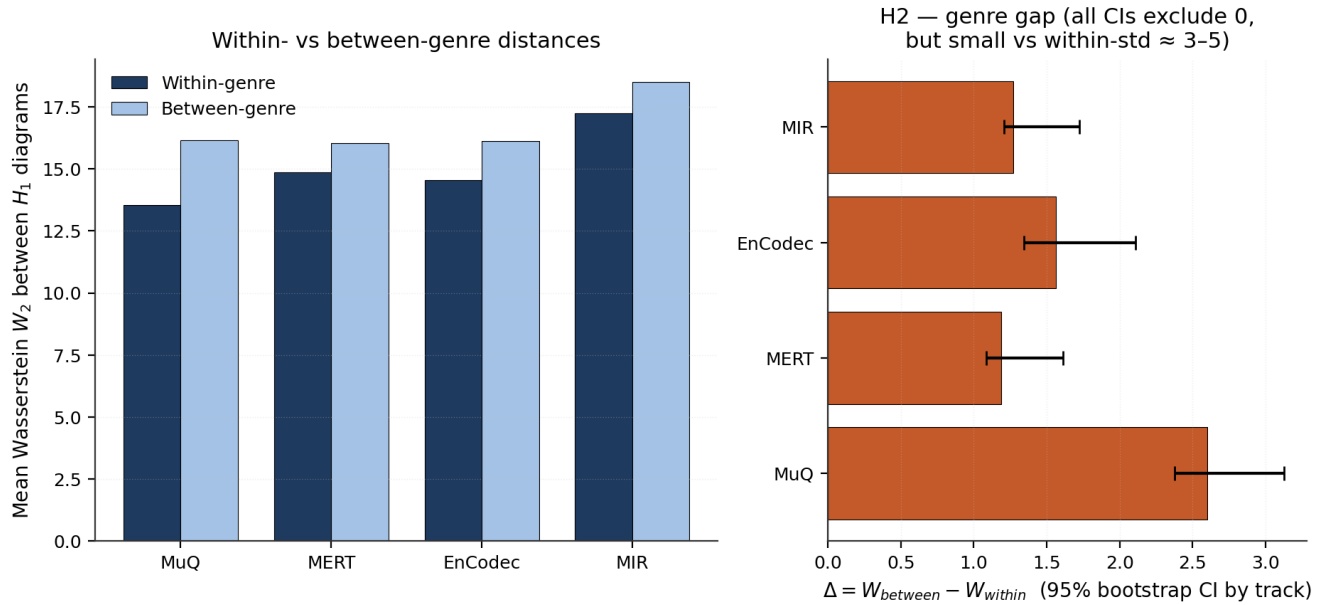


Figure 6 – Within-genre and between-genre Wasserstein distances.

All confidence intervals exclude zero, so the hypothesis is formally supported. However, the effect size is small: the gap is only about 0.4–0.9 within-genre standard deviations (within-std $\approx$ 2.9–4.9). The between-genre difference is drowned by within-genre variation.

Conclusion: H2 is weakly supported: statistically significant but not practically strong. Topological shape is only weakly genre-specific.

6.3 H3 — Nonlinear Temporal Structure

Comparison of real data with phase surrogates by maximum H_1 persistence is shown in Table 4 and Figure 7.

Table 4 – Real vs. phase surrogate comparison by maximum H_1 persistence

Channel	Real	Phase	Rank-biserial
MERT	0.465	0.324	0.972
MuQ	0.326	0.233	0.995
EnCodec	0.374	0.266	0.925
MIR	1.595	1.248	0.901

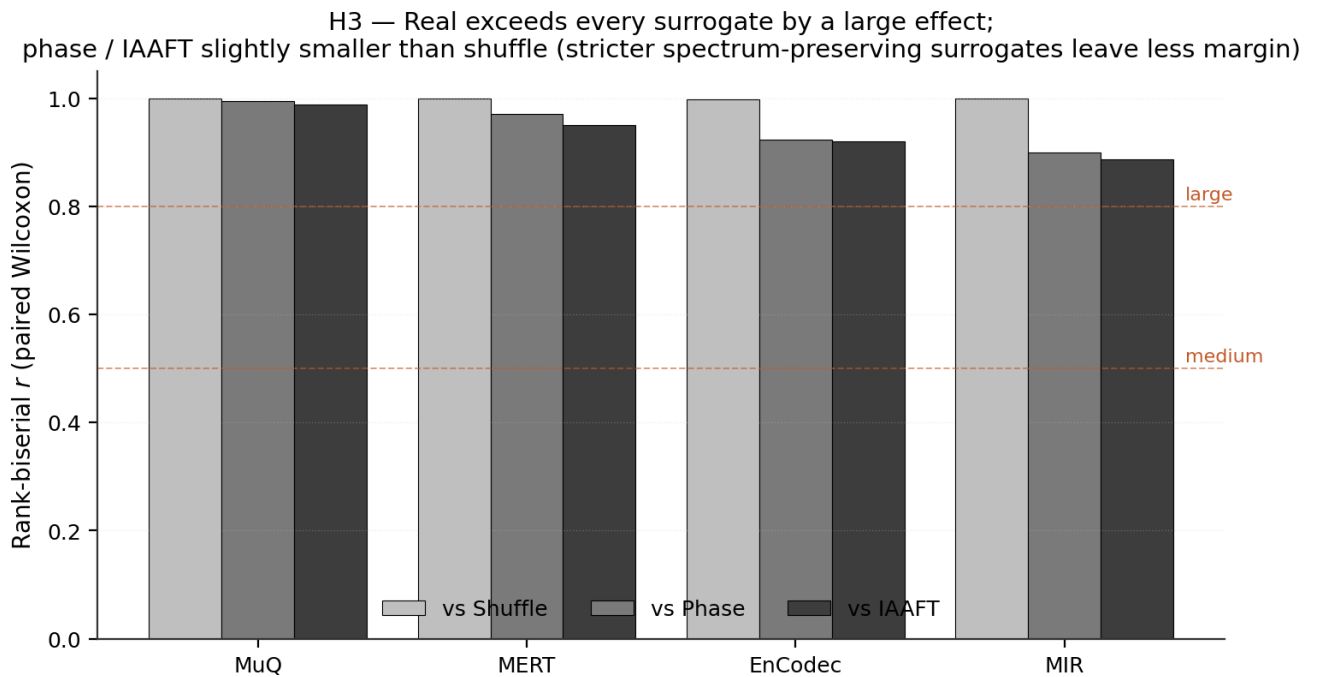


Figure 7 – Comparison of real data with phase surrogates by maximum H_1 persistence.

In all four spaces, real trajectories exceed phase surrogates with large effect sizes. This indicates a component of recurrent geometry that is not explained by the chosen linear-spectral surrogate. Interpreting this as nonlinear structure is plausible, but it is not a strict proof of nonlinearity in the dynamical-systems sense.

Conclusion (careful H3 formulation): the experiment does not test “musicality” as such; that would require comparison to non-musical audio. It tests whether observed topology exceeds a concrete phase surrogate. The answer is positive for all four representations, though the effect magnitude differs: stronger for MuQ/MERT and lower for EnCodec/MIR.

6.4 H4 — Representation Consistency

The Mantel test with permutations between pairwise Wasserstein distance matrices is shown in Table 5.

Table 5 – Mantel test results for cross-representation consistency

Pair	r	p	95% CI
MERT–MuQ	0.279	0.0001	[0.245, 0.319]
MERT–EnCodec	0.005	0.7553	[−0.023, 0.046]
MuQ–EnCodec	0.064	0.0001	[0.037, 0.105]

MERT–MuQ has a weak correlation ($r = 0.28$), plausibly because both are SSL music-related models. MuQ–EnCodec shows a tiny correlation ($r = 0.064$) with a small p -value: a clear example of significance driven by sample size rather than effect magnitude. MERT–EnCodec is not significant.

Conclusion: H4 is not supported. Topological organization depends strongly on representation. Different embedding spaces do not share a common topological structure.

6.5 H5 — Genre Classification

Classification results for the MuQ channel are shown in Figure 8 (chance ≈ 0.10).

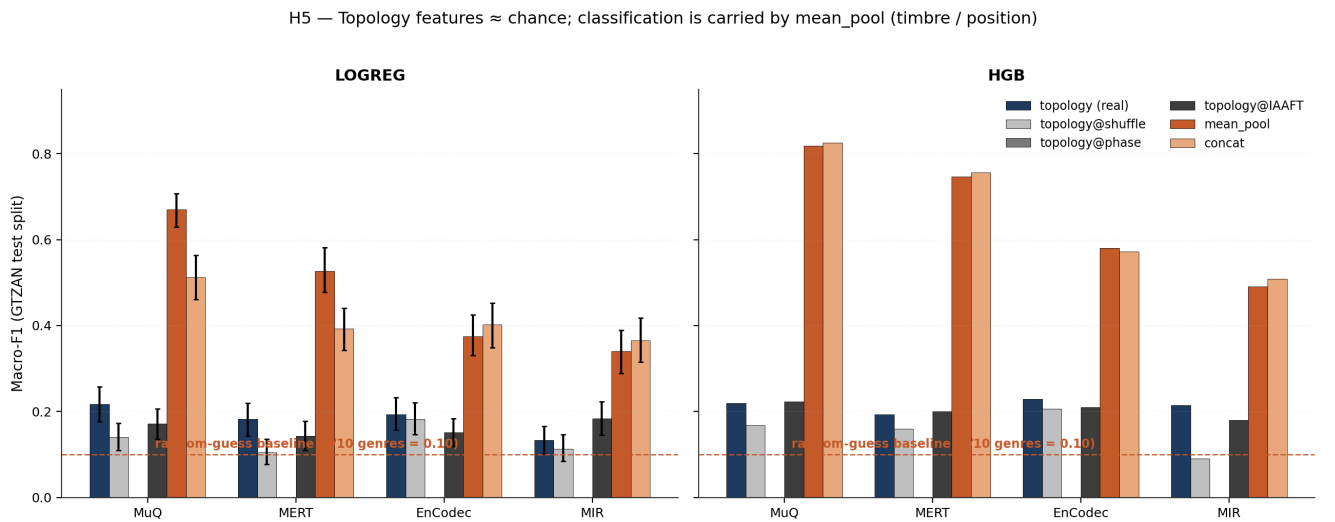


Figure 8 – Genre classification macro-F1 across features and models.

Three independent arguments against H5:

1. **Absolute gap:** topology ≈ 0.22 versus baseline ≈ 0.82 . Topology is near chance; the baseline is much better.
2. **Classifier loophole closed:** HGB improves the baseline (0.82 vs. 0.67) but does not make topological features practically useful (MuQ: 0.219 vs. 0.217 for LogReg; the phase-surrogate HGB score is even higher than real topology). Thus the issue is not simply a weak linear model.
3. **Concatenation:** for HGB, concat $\approx$ baseline (0.825 vs. 0.818); topology adds no value. For LogReg, it even hurts (0.512 vs. 0.670).

The applicability map by channel is summarized in Table 6.

Table 6 – Classification macro-F1 by channel (LogReg unless otherwise noted)

Channel	F1 real	F1 shuffle	F1 phase	Signal above controls
MuQ	0.217	0.141	0.214	above shuffle, not above phase
EnCodec	0.193	0.183	0.170	CIs overlap
MERT	0.182	0.104	0.152	weak, near chance
MIR	0.133	0.113	0.241	phase above real

Conclusion: H5 is not supported. Genre is encoded mainly in timbre and average position in feature space, not in trajectory shape.

6.6 H-loop — Chromatic Correspondence of Loops to Repeated Sections

Progression as methodology was corrected is shown in Table 7 and Figure 9.

Table 7 – H-loop: fraction of significant tracks as methodology was corrected

Method	Significant tracks
Cocycle + unpaired Mann–Whitney	~55%
Cocycle + permutation test	~43% raw, 35% FDR
Dionysus cycle + permutation + FDR	34% raw, 27% (52/192) FDR

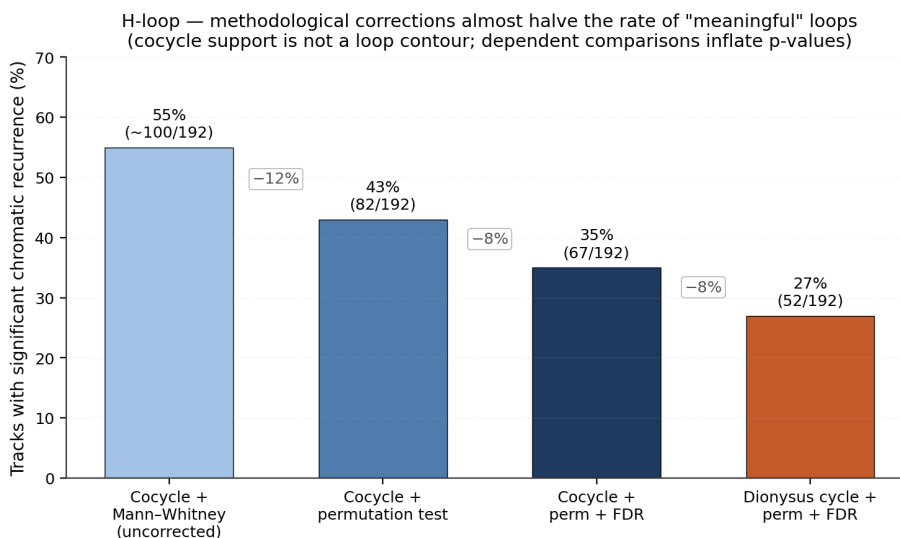


Figure 9 – Progression of H-loop significant track fraction as methodology was corrected.

The original estimate (~55%) was almost two times too high due to two artifacts: (1) cocycle support is not a loop contour (median 85 points out of 300 versus median 36 for a homological cycle, as compared in Figure 10), and (2) the unpaired test treated dependent comparisons as independent observations.

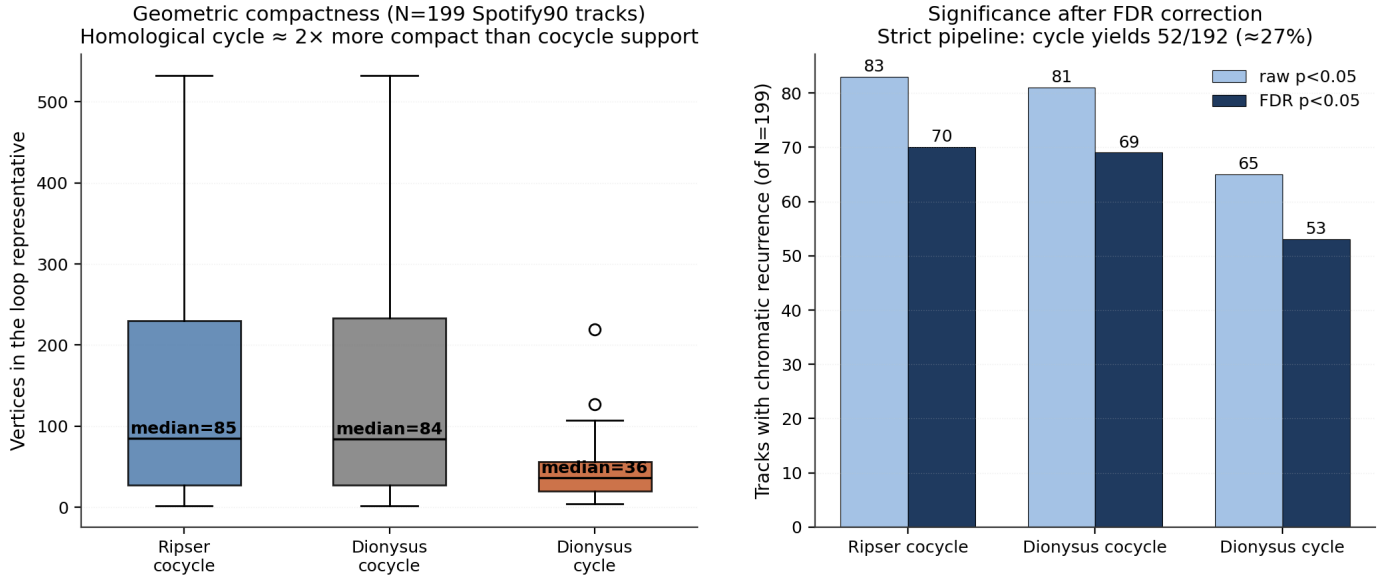


Figure 10 – Comparison of cycle and cocycle supports on empirical Takens embeddings of audio.

The honest estimate (chroma permutation test on a Dionysus cycle, FDR) is that about 27% of tracks (52/192 under master FDR; a direct cycle-table flag gives 53/192 because of one borderline track near the 0.05 threshold) show increased chromatic similarity among cycle vertices. Among tracks in the top quartile of persistence, the rate is about 45%, but the dependence on persistence is weak.

Conclusion: H-loop is weakly and partially supported. For a minority of tracks, the homological cycle is localized on harmonically recurring sections. The correct formulation is: for about 27% of tracks, the cycle detects chromatic recurrence.

6.7 Robustness Analysis

Systematic ablation on a stratified subset (~ 100 tracks, 10 per genre) is summarized in Table 8 and Figure 11.

Table 8 – Robustness analysis: sensitivity to pipeline parameters

Factor	Conclusion
Random split (5 seeds)	Real-vs-shuffle effect = 1.0 ± 0.0 ; gap varies but sign is stable
PCA dimension (16–128)	Effect is stable; gap grows with number of components
Normalization (4 variants)	Loop robustness remains; StandardScaler $\approx$ RobustScaler (gap 3.43 vs 3.44)
Window stride ($s = w/2, w/4, w/8$)	Gap decreases with stride; sign remains stable
Window length (0.32–2.56 s)	Gap from ~ 6 (0.3 s) to < 1 (2.5 s); channel order preserved
Resampling method (Fourier vs linear)	Qualitative picture unchanged
MERT/EnCodec resampling confound	Without resampling, gap does not increase: MERT 0.60 (base 0.76), EnCodec 1.47 (base 1.49); MuQ advantage is model-related
Persistence metric	Max-persistence is stable; aggregate loop counts are unstable because shuffle creates many short-lived loops
Distance concentration	Takens CV ≈ 0.10 compresses absolute persistence but does not remove real-vs-control separation

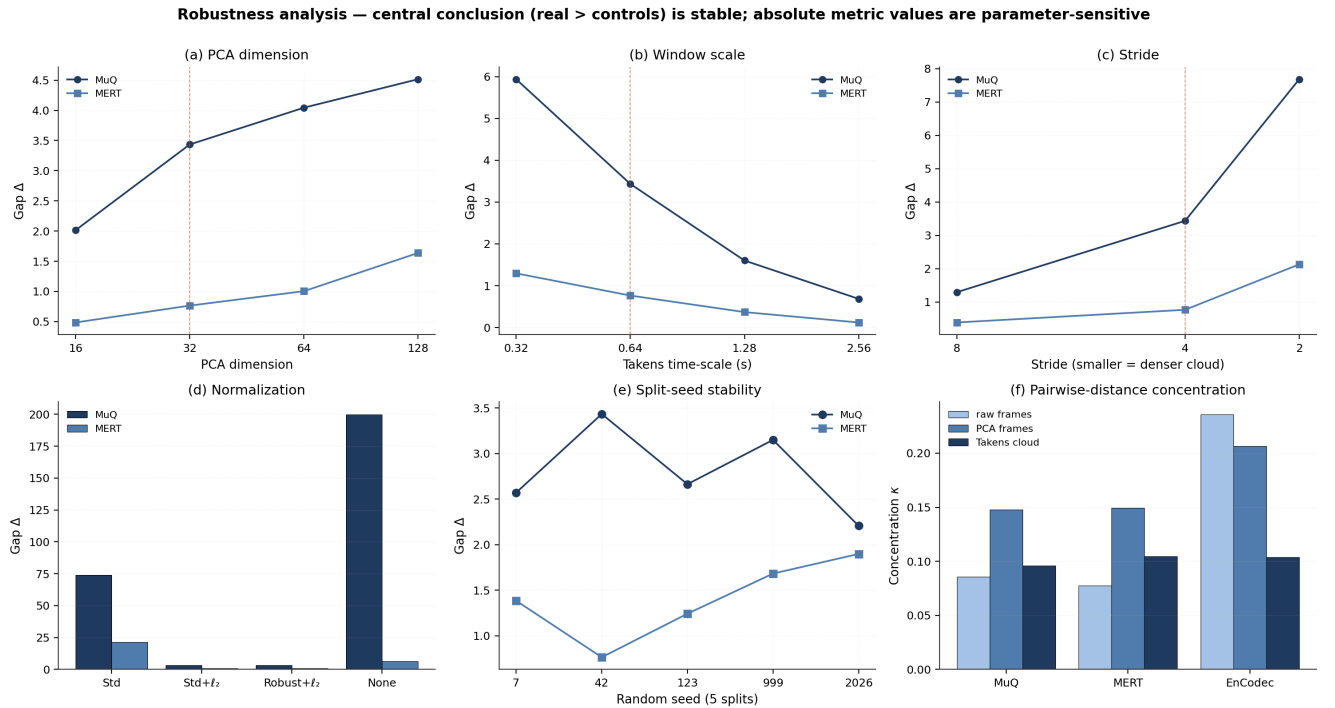


Figure 11 – Robustness analysis panel across various factors.

The distance-concentration row should be interpreted as a diagnostic rather than a pass/fail test: in a high-dimensional Takens space, pairwise distances become compressed, so absolute persistence values are parameter-sensitive. The relevant observation is that the separation from surrogate controls remains.

The persistence-metric ablation is also important. Max-persistence tracks the most prominent loop and remains stable across controls. Aggregate metrics such as total loop count or total persistence

are less reliable because shuffle can generate many short-lived noisy cycles; therefore they can change sign even when the dominant-loop conclusion is stable. This stability is illustrated in Figure 12.

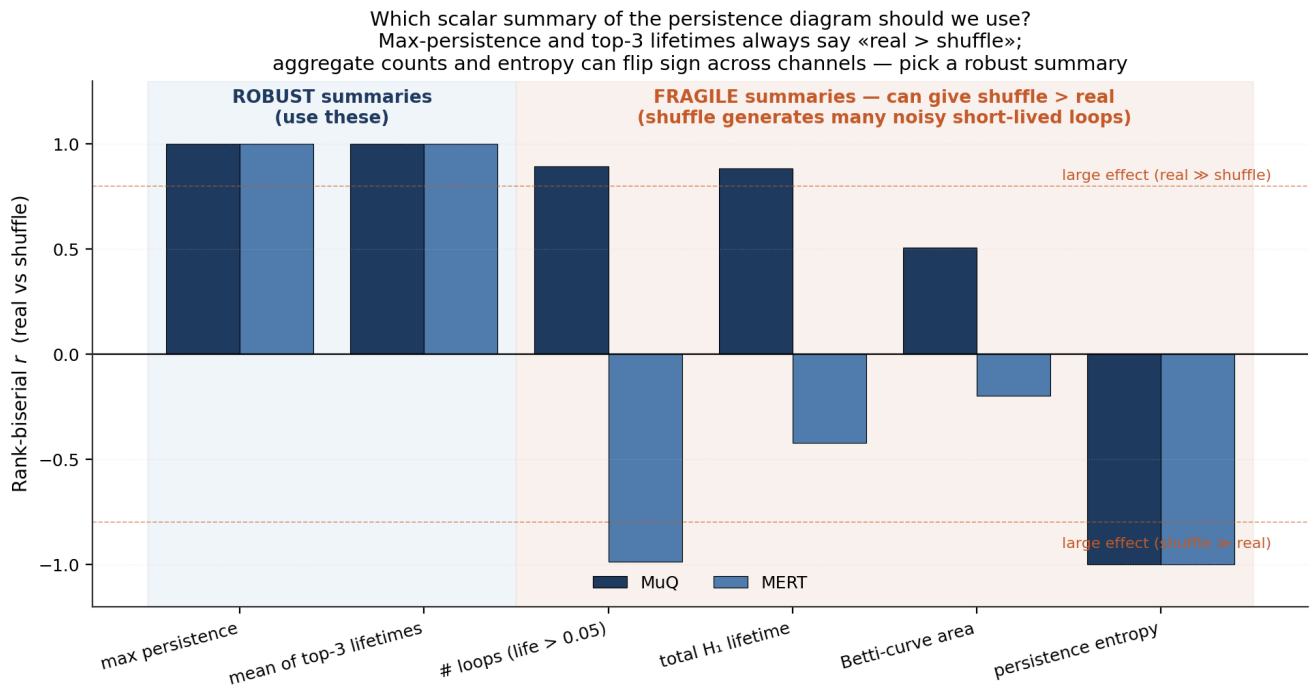


Figure 12 – Ablation of various persistence metrics and their stability.

Conclusion: the central result (real loops > controls) is robust. Absolute metric values are parameter-sensitive, but their sign and channel order are not.

6.8 Multiple-Comparison Control

In any study conducting numerous statistical tests across different models, channels, and individual tracks, there is a high risk of inflated Type I error (false positives). To rigorously control for this, the Benjamini–Hochberg False Discovery Rate (FDR) correction was applied. Rather than applying a single blanket correction to all tests—which would overly penalize independent hypotheses—the correction was applied within logically distinct “test families” (see Table 9).

Table 9 – FDR correction results by test family

Family	Tests	Significant after BH
controls_shuffle	4	4 (all)
controls_persistence	8	8 (all)
mantel	3	2 (MERT–MuQ, MuQ–EnCodec)
chroma_loop_cycle	192	52 (27%)

Explanation of test families:

- **controls_shuffle and controls_persistence:** These families cover the tests for H1 and H3, comparing real persistence against Shuffle, Phase, and IAAFT surrogates across the four rep-

resentation channels. All tests survived FDR control, confirming that the topological reality of loops is a robust, system-wide phenomenon rather than a statistical fluke in a single channel.

- **mantel:** This family controls the pairwise Mantel tests evaluating cross-representation consistency (H4). The weak consistency between MERT/MuQ and MuQ/EnCodec survived, while MERT/EnCodec remained non-significant, reinforcing the conclusion that different models induce fundamentally different topological geometries.
- **chroma_loop_cycle:** This is the most critical correction. The H-loop hypothesis required running 192 independent permutation tests (one for each track). Without correction, 65 tracks ($\sim 34\%$) showed raw $p < 0.05$. After applying the strict Benjamini–Hochberg FDR control to this family of 192 tests, 52 tracks (27%) remained significant. This proves that for a distinct minority of tracks, the alignment of the topological cycle with chromatic musical recurrence is a genuine effect, not an artifact of testing 192 songs simultaneously.

Conclusion: The survival of the core findings through strict FDR control confirms that the methodological framework successfully filters out random noise. The results regarding real topological loops, representation-specific geometry, and track-level chromatic correspondence are statistically robust.

6.9 Popularity and Topological Findings

As an additional exploratory analysis, Spotify popularity was tested against topological loop statistics for the 198 matched tracks. The matched subset covers only four genres (electronic, hip-hop, pop, reggae), all with popularity from 56 to 99 (median 83). Therefore, this analysis cannot answer whether popular music differs from unpopular music; it only checks whether popularity variation inside a top-track subset aligns with topology. Summary statistics are shown in Table 10, and the main Spearman correlations are shown in Table 11 (see also Figure 13).

Table 10 – Popularity subset summary

Space	n	Pop. median	H-loop sig.	Mean max pers.	Mean chroma excess
MuQ	198	83	50.0%	0.353	0.028
MIR	198	83	76.8%	1.636	0.049

Table 11 – Spearman correlations between Spotify popularity and topological metrics

Space	Metric	ρ	p	Genre-centered ρ	Genre-centered p
MuQ	max H_1 persistence	−0.030	0.677	0.052	0.467
MuQ	chroma excess	−0.107	0.141	−0.068	0.351
MIR	max H_1 persistence	−0.042	0.562	−0.053	0.457
MIR	chroma excess	0.017	0.812	−0.120	0.098

No robust association survives this exploratory analysis. Maximum persistence, cycle size, and temporal span are essentially uncorrelated with popularity in both spaces. Chroma excess shows a weak

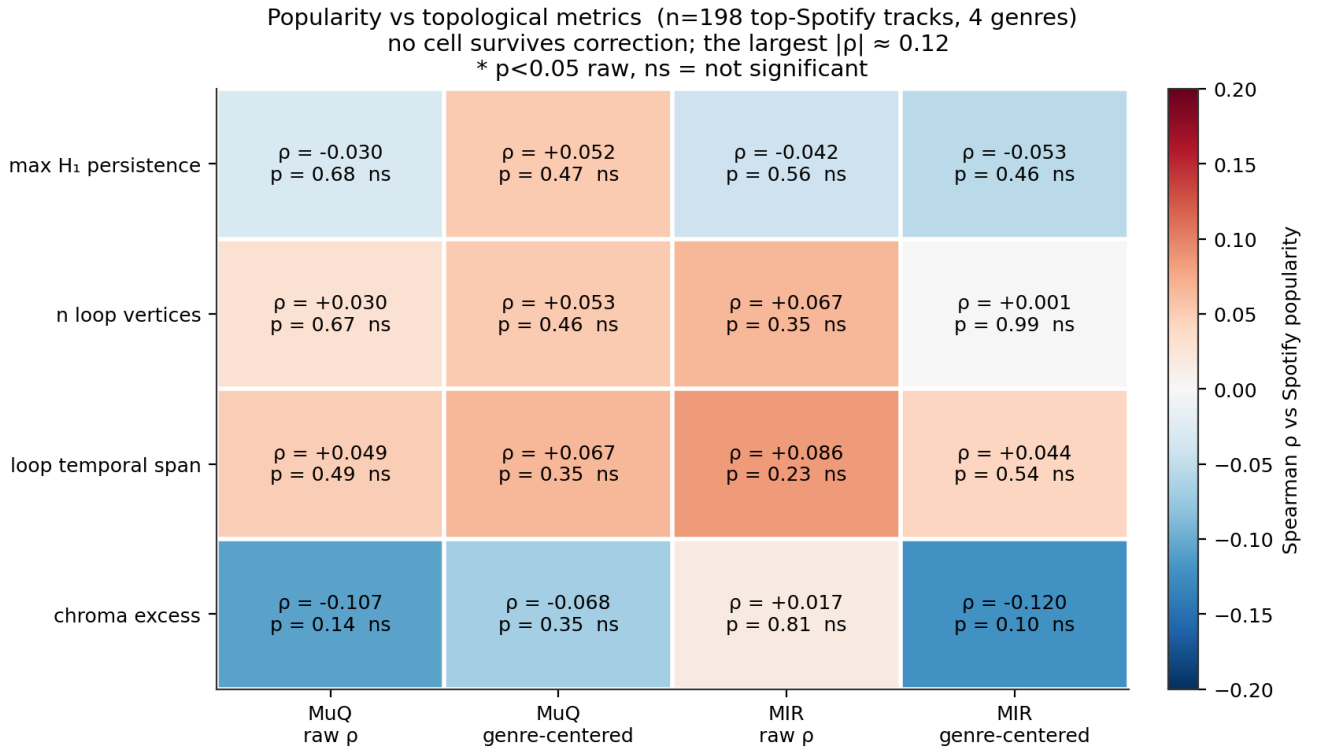


Figure 13 – Correlations between Spotify popularity and topological metrics.

negative tendency after genre control in MIR and in high-vs-low comparisons, but it is not significant after multiple-comparison correction and should not be interpreted as evidence of a real popularity effect. The most defensible conclusion is negative: in this top-track subset, popularity does not appear to be explained by stronger topological loops or by clearer chromatic recurrence.

7 Discussion

The goal of the work was a rigorous audit of the applicability of TDA to trajectories of neural audio embeddings, not the development of a new music method. The results delineate where topological organization of trajectories is meaningful and where it is not.

7.1 What Trajectory Topology Captures

The most robust result is that H_1 loops in trajectories reflect real recurrent structure rather than computational artifacts. This conclusion is supported by the surrogate ladder testing order, linear-spectral smoothness, and value distribution. Across all four spaces, observed persistence exceeds all three surrogates with large effect sizes (rank-biserial ≈ 0.9 – 1.0). This means:

- **Temporal recurrence:** the signal returns to nearby states due to internal trajectory dynamics, not by chance.
- **A component beyond the phase surrogate:** real data exceed a linear-spectral control in all four channels.
- **Sensitivity to temporal order:** shuffle destroys loops with maximal effect, so frame order, not only frame distribution, is a source of geometry.

7.2 What Trajectory Topology Does Not Capture

When the question shifts from “is the structure real?” to “does it carry semantic information?”, the answer is negative along several independent lines:

- **Genre membership:** topological features reach only $F1 \approx 0.22$ – 0.23 versus 0.82 for `mean_pool`. The “weak classifier” loophole is closed: genre signal is absent from topology, not merely inaccessible to a linear model.
- **Timbral identity:** genre is encoded in average position in feature space (`mean_pool`), not in path geometry.
- **Cross-model invariance:** topological organization depends strongly on the embedding model. Distance-matrix correlations are weak even between SSL models ($r = 0.28$) and negligible compared to the codec.
- **Musical repetitions in most tracks:** only about 27% of tracks show chromatic recurrence along the homological cycle; for the rest, the loop exists geometrically but is not concentrated on repeated sections.

7.3 Methodological Insight: Strict Surrogates Matter More Than Shuffle

The difference between shuffle and stricter smooth surrogates is crucial. Real trajectories exceed shuffle almost perfectly in every channel, but in downstream classification the advantage of topology over phase/IAAFT is small or absent. Thus, shuffle only answers whether temporal order matters; it does not establish musical or genre-level content. Phase and IAAFT controls are therefore necessary: they separate crude order effects from structure beyond smooth correlated processes.

7.4 Limitations

Several pipeline choices must be explicitly recognized as limitations.

Takens-induced recurrence. Delay embeddings naturally tend to generate loop-like geometry for periodic and quasiperiodic signals. The mere existence of H_1 loops is not evidence by itself; evidential value arises only through comparison with surrogates.

Dependence on PCA. Explained variance in 32 components differs substantially: EnCodec $\approx 84\%$, MERT $\approx 36\%$, MuQ $\approx 22\%$. Point clouds live in spaces with different information content, complicating direct comparison of absolute persistence values.

Unit-sphere normalization is applied before PCA and is a deliberate choice to analyze shape rather than energy. This is a substantive decision affecting cloud geometry, not a neutral technical detail.

Heuristic embedding parameters (window = 16, stride = 4). Robustness to variation was tested by sensitivity analysis, but optimality is not guaranteed.

Componentwise IAAFT does not preserve cross-spectral dependencies between components, so the surrogate ladder is not strictly nested.

H-loop null model. The chroma test compares cycle vertices with random time points of the same size and uses only pairs separated by at least 5 seconds. This reduces local autocorrelation effects but does not fully control the temporal geometry of the cycle.

Popularity analysis is strongly sample-biased: it uses only cached Spotify90 embeddings for top tracks from four genres. The absence of correlation should be read narrowly.

GTZAN defects: known label errors and duplicates introduce noise in genre labels [5].

7.5 Methodological Lessons

Diagram vectorization is critical. The same topological signal looked absent under persistence images, which degenerated for near-diagonal loops, but became visible under Betti curves. The fragility of the “TDA $\rightarrow$ ML” bridge becomes apparent only when vectorization is diagnosed.

Significance is not usefulness. With 999 tracks, a tiny gap becomes statistically significant due to sample size. It does not imply predictive power. Downstream metrics with bootstrap confidence intervals are more informative than a single p -value.

Cocycle vs. cycle. The standard Ripser output is a cocycle; its vertex support is not a geometric loop contour. For temporal interpretation, a homological representative cycle is more appropriate and is about twice as compact in this dataset. Ignoring this distinction nearly doubles the estimated fraction of “meaningful” loops.

8 Conclusion

This work presents a rigorous, controlled, and reproducible audit of applying TDA to trajectories of musical audio embeddings. The method reliably detects temporal recurrence structure, but it does not carry useful genre information, is not consistent across embedding models, and requires caution when selecting geometric representatives for loop interpretation.

The hypothesis status is summarized in Table 12.

Table 12 – Summary of hypothesis testing

Hypothesis	Status
H1 — Reality of loops	Strongly supported. Rank-biserial ≈ 1.0 , all surrogates, all channels
H2 — Genre specificity	Weakly supported. CIs exclude 0, but effect is small (gap ≈ 0.4 – 0.9 std)
H3 — Nonlinear structure	Supported with caveat. Real $>$ phase in all four channels; a component beyond the chosen linear-spectral surrogate, not a strict proof of nonlinearity
H4 — Cross-representation consistency	Not supported. Strong representation dependence; $r \leq 0.28$
H5 — Genre classification	Not supported. Best topology F1 ≈ 0.23 near chance vs. 0.82 for mean_pool
H-loop — Chromatic correspondence	Weakly and partially supported. 27% of tracks; chromatic recurrence, not “loop = repetition”
Popularity vs. topology	Not supported in the exploratory subset. On 198 cached Spotify90 top tracks, popularity is not robustly correlated with persistence, cycle size, cycle span, or chroma excess

The main contribution is not a new tool, but a **map of applicability**: a clear separation between what topological organization of embedding trajectories can measure (temporal recurrence, sensitivity to order, and a component beyond a linear-spectral surrogate) and what it cannot measure (genre semantics, timbral identity, and model-invariant properties of music). The negative results are central: finding a sharp separation between trajectory topology and genre semantics is an informative result, not a failure of the method.

9 Future Work

Semantically compatible tasks. Since topology captures recurrence and movement structure rather than timbre, promising tasks are those where temporal trajectory shape is inherently informative: structural segmentation, recognition of musical form (verse/chorus/bridge), and cover-song identification (timbre changes but structural skeleton remains). A positive result on such tasks, combined with a negative result on genre, would support the proposed explanation separating “shape” from “position.”

Cycle as a repetition detector. A homological representative cycle extracted through Dionysus can be compared with existing segmentation methods based on self-similarity matrices and evaluated as a standalone repetition detector. The cocycle/cycle distinction identified in this work directly informs such research.

Embedding interpretability. Weak topological consistency across spaces (H4) raises the question of how representation topology relates to training objectives and model data. This connects the work to interpretability research on neural audio embeddings.

Methodological refinements. (1) Systematic selection of delay-embedding parameters, for example via mutual information [13], instead of heuristics. (2) Multivariate and pseudoperiodic surrogates as finer null models. (3) Higher-dimensional homology (H_2 , H_3), whose musical interpretation remains unexplored. (4) A stricter H-loop null model using matched random sets by temporal-distance distribution or block/circular permutation.

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